

Getting Started with SharinPix Einstein

Requirement

- You need to install the latest version of SharinPix from the AppExchange
- You need to sign up for a Einstein Vision API account from <https://api.einstein.ai/signup>

Setup

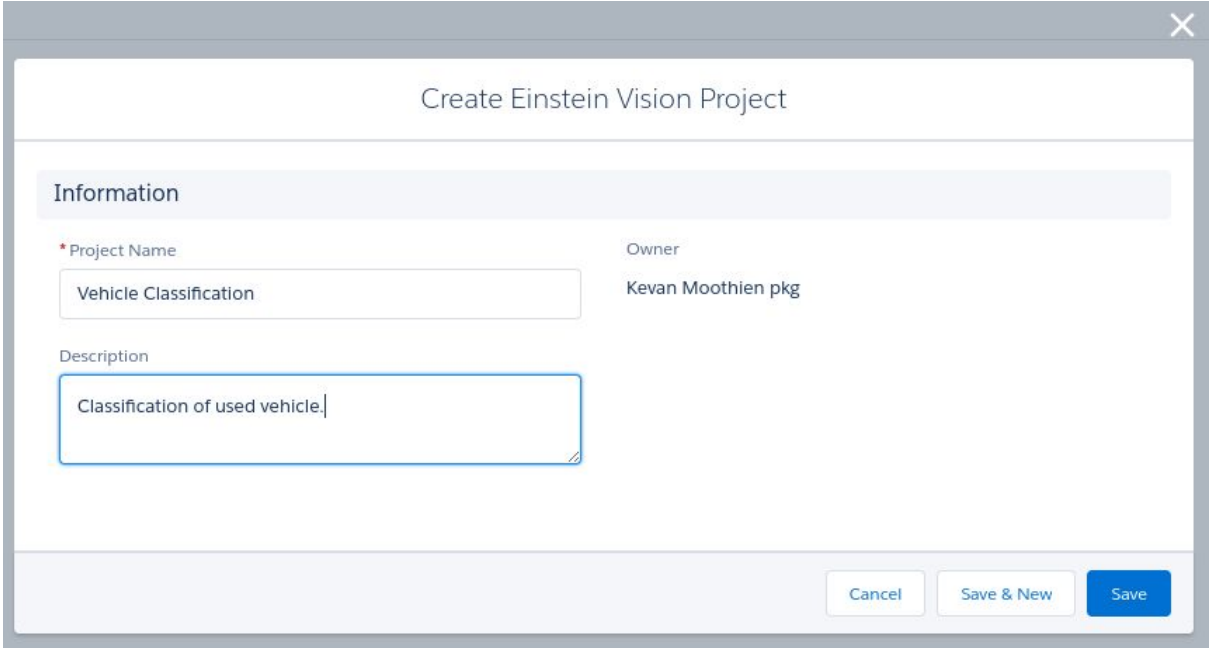
After you have sign up for the Einstein Vision API account, to just need to contact us and send the Email address and the file “einstein_platform.pem” given to you.

Getting started

Einstein Vision allow you to predict what is an image against a specific set of images.

SharinPix allows you to train your own version of Einstein from your images. To do so, click the Einstein Vision Project tab and click on the button New.

Fill in the fields and save.



Create Einstein Vision Project

Information

* Project Name
Vehicle Classification

Owner
Kevan Moothien pkg

Description
Classification of used vehicle.

Cancel Save & New Save

Now that your project is ready, you need to create the different labels that you want the image to be recognized as.

For our demo purpose, we will use the following:

- Car
- Truck
- Motorcycle
- Bus

On the Related list, click on **New** from the Einstein Vision Label Related list and create all the desired labels.

RELATED		DETAILS
Einstein Vision Labels (4)		New
LABEL NAME	DESCRIPTION	
Car		
Truck		
Motorcycle		
Bus		
		View All

Go to details, add your images and tag them according to the labels you want.

EINSTEIN VISION PROJECT

Vehicle Classification

RELATED

DETAILS

Project Name

Vehicle Classification

Owner

Kevan Moothien pkg

Description

Classification of used vehicle.

Images

50 Images selected

Tags

Bus

Car

Motorcycle

Truck

Select all

Cancel























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Once you have completed tagging all the images with the appropriate label, you can now create you Einstein Vision Dataset.

This dataset is a copy of the Einstein Vision Project at the time when the dataset was created and it is frozen. It is this dataset that you will use to train the model against which you will do the prediction.

Click on the related list for the Einstein Vision Project and click New next to the Einstein Vision Dataset.

×

Create Einstein Dataset

Information

* Name

Vehicle Dataset 1

* Einstein Vision Project

Vehicle Classification

×

Description

First dataset for vehicle classification

Cancel

Save & New

Save

Open the dataset and you will see all your images which be displayed.

Vehicle Dataset 1

RELATED

DETAILS

Name

Vehicle Dataset 1

Einstein Vision Project

Vehicle Classification

Description

First dataset for vehicle classification

Album

Tags

Sort by

✓



Motorcycle

✓



Motorcycle

✓



Motorcycle

✓



Motorcycle

✓



Motorcycle

✓



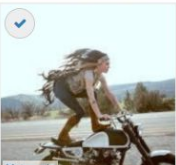
Motorcycle

✓



Motorcycle

✓



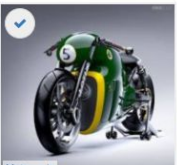
Motorcycle

✓



Motorcycle

✓



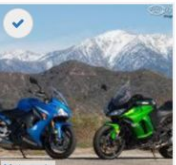
Motorcycle

✓



Motorcycle

✓



Motorcycle

←

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...

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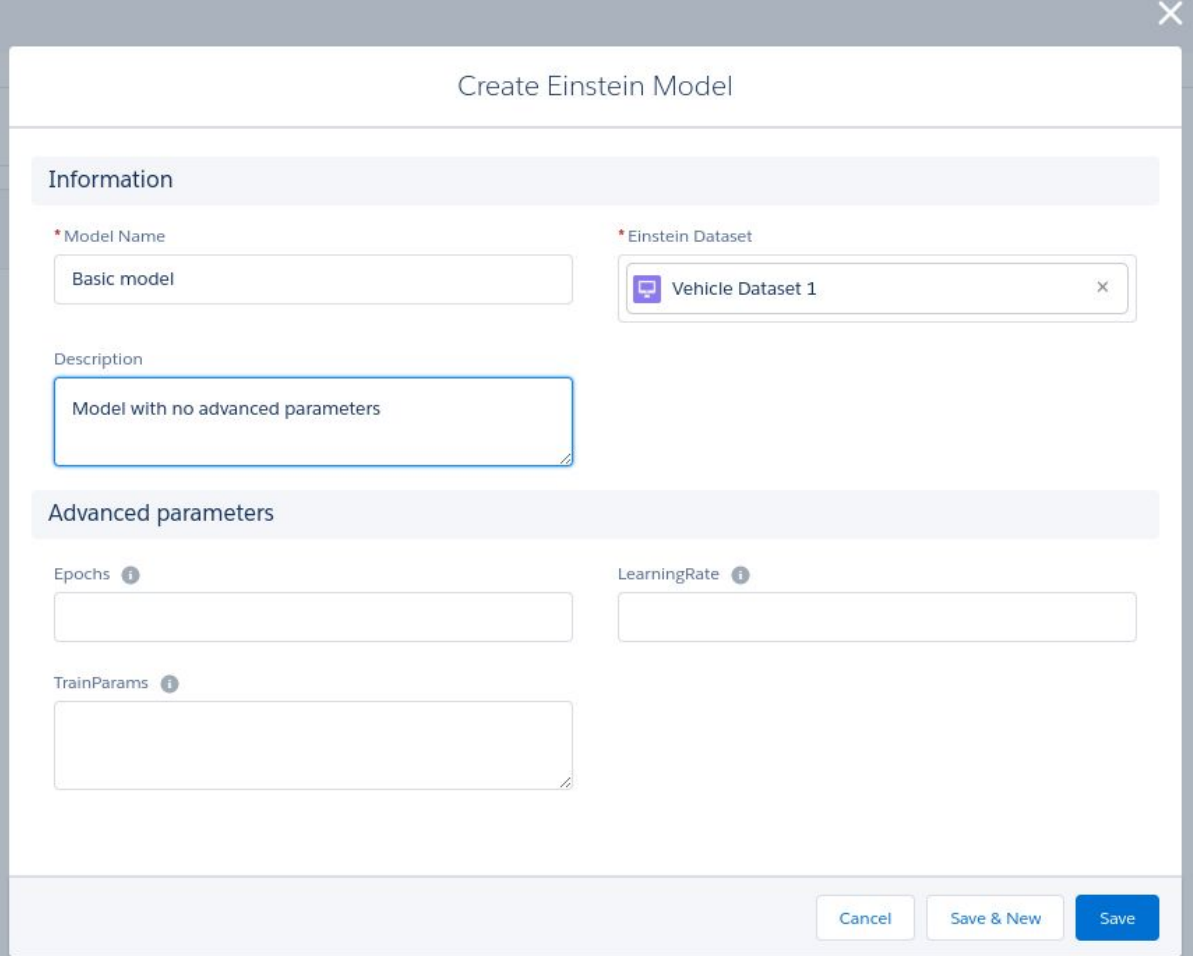
...

19

→

Now that we have all our images uploaded and correctly labeled, we can proceed with the training process.

For this, we need to create the Einstein Model. Click on the related list and click New next to the Einstein Model related list.



The screenshot shows a 'Create Einstein Model' dialog box with a close button (X) in the top right corner. The dialog is divided into two main sections: 'Information' and 'Advanced parameters'.

Information Section:

- * Model Name:** A text input field containing 'Basic model'.
- * Einstein Dataset:** A dropdown menu showing 'Vehicle Dataset 1' with a close button (X) on the right.
- Description:** A text input field containing 'Model with no advanced parameters'.

Advanced parameters Section:

- Epochs:** A text input field with an information icon (i) to its right.
- LearningRate:** A text input field with an information icon (i) to its right.
- TrainParams:** A text input field with an information icon (i) to its right.

At the bottom right of the dialog, there are three buttons: 'Cancel', 'Save & New', and 'Save'.

Guide for using the advanced parameters can be found here:

<https://metamind.readme.io/docs/train-a-dataset>

Open the Einstein Model record

 **EINSTEIN MODEL**
Basic model

RELATED

DETAILS

Model Name

Basic model

Einstein Dataset

Vehicle Dataset 1

Description

Model with no advanced parameters

Advanced parameters

Epochs

LearningRate

TrainParams

Model status

17%

Model status

Model Id: GMC64S6LIB45HF3N6OKL7J54BE

Status: RUNNING

Number of labels:

Number of images:

The training process may take a while depending on the amount of images on your dataset. It also depends on the Einstein Vision server on Salesforce on which we do not have any control.

When the training is completed, you can on the spot check whether your model was trained correctly.

Insert images in the album which is below the status section. Open the image in large and you will see a spinner which will appear. This means that prediction is in progress and the result will be displayed soon.

100%

Model status

Model Id: GMC64S6LIB45HF3N6OKL7J54BE

Status: SUCCEEDED

Number of labels: 4

Number of images: 219

Prediction status

Truck 1

Bus 2.335513e-9

Car 5.064336e-11

Motorcycle 1.0086336e-11



The prediction is a probability from 0 - 1.

Usage

We talked about setup and how to train a dataset to generate a model. Now we will discuss how to leverage those information and apply it to your business rules.

To predict an image and retrieving the prediction result are done in two phases.

1. Call the SharinPix api to ask for a prediction. A sample code below:

```
string imageId = '79b86fd9-5f0c-4ac2-a4e7-2d875f254a2a';
string modelId = 'a04240000Ns4HdAAJ'; //Id of the Einstein Model on salesforce
map<string, string> payload = new map<string, string>{
    'imageId' => imageId,
    'modelId' => modelId
};
sharinpix.Client clientInstance = sharinpix.Client.getInstance();
object response = clientInstance.post('/einstein_prediction',
    Blob.valueOf(JSON.serialize(payload)),
    new map<string, object>{'einstein'=>true});
```

2. For retrieving the prediction, you have two options,

- a. Keep running the above code until you have a response with a probability in it.
- b. Or let SharinPix notify you via **Webhook**

Webhook

Here how to configure a classic webhook which will post the information on a Salesforce Site.

https://docs.google.com/document/d/1kL2I9wJSHGL9H5s89K7r92xAtoANLuj_w82ltadfFTA/edit#heading=h.l9wwwfabrc06

But in our case we will another type of webhook which will execute a static method on your Salesforce organization.

To configure it, you first need to create an apex class and add a static method on your Saelsforce environment.

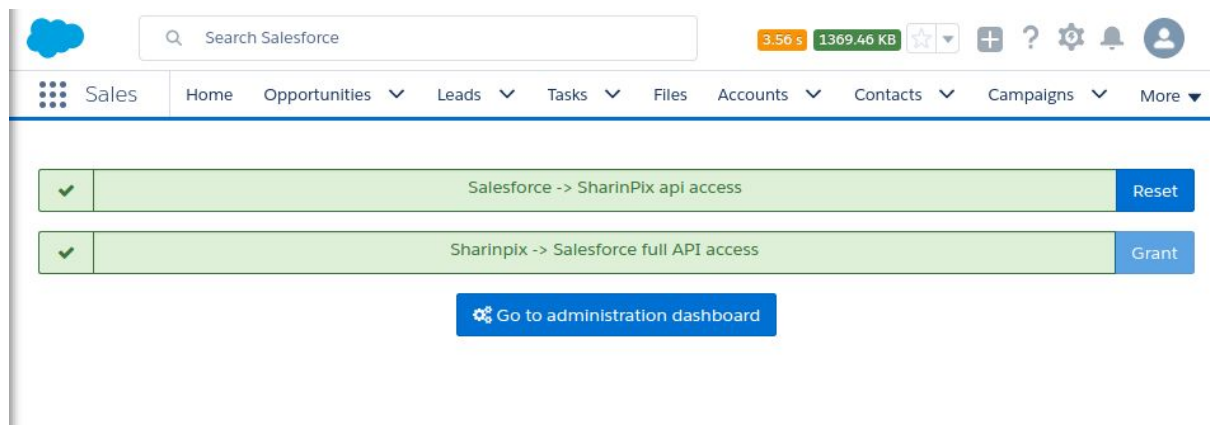
Sample:

```
public class Webhook {  
    public static void perform(string payload, string webhookInformation){  
        // your business logic here  
    }  
}
```

payload is the a json string containing the prediction value.

webhookInformation contains information about the webhook such as the event type.

Now check whether you have already granted the Salesforce Full API access to SharinPix. To do so, go to the SharinPix Settings tab and check. If not, click on Grant to start to authenticate.



Next, go to the SharinPix Administration dashboard, click on the menu **Webhooks** and **New Webhook**. Select Action type **apex_method** and insert your class name and method name and select **New einstein prediction** as event type.

ADMIN / WEBHOOKS /

New Webhook

Organization*	SharinPIX-00D24000000cfi0EAA (906f1e50) ▼
Action type	apex_method ▼
Class name	Webhook
Method name	perform

☐ New image

☐ New publication

☐ New provider reply

☐ New tag image

☐ Delete image

☐ Delete tag image

☒ New einstein prediction

Create Webhook

Cancel

Click on **Create Webhook** to save.

Whenever a prediction call is made, this webhook will be executed as soon as a response is received from Einstein.